EPS tap FeatherWing: input front end for the Honda Insight EPS logger

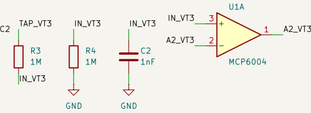
Taps connect at the EPS control unit connector B (14P green) and A2 (+B). Every car-facing line goes through >=100 kOhm (1 MOhm for the torque sensor) before anything else, so a fault on the logger side can't load the EPS. Fit the plug with the ignition OFF. See ../../BOM.md. JP1–JP14 (open as shipped) short the series resistors: bridge them only when the resistors are fitted at the probe ends instead.

Analog inputs, buffered (1 MOhm + divider + 1 nF, MCP6024 follower on 3V3)

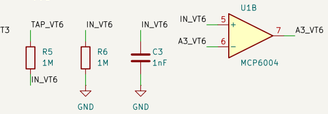
VCC2



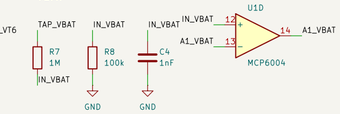
VT3

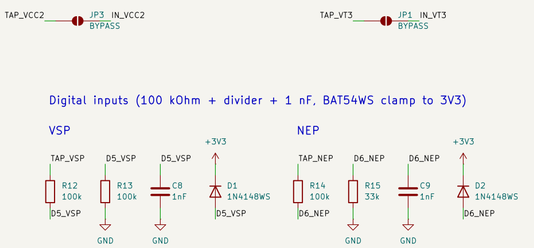


VT6



VBAT



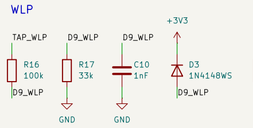


Digital inputs (100 kOhm + divider + 1 nF, BAT54WS clamp to 3V3)

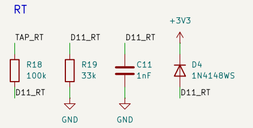
VSP

NEP

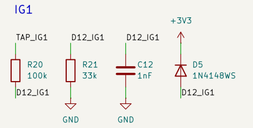
WLP



RT

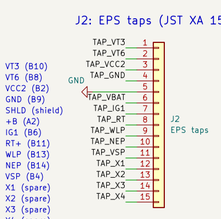


IG1

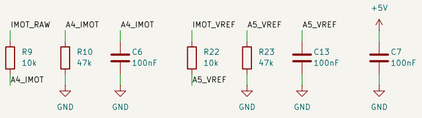




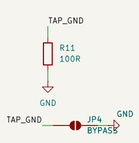
J2: EPS taps (JST XA 15-way, 2.5 mm, locking)



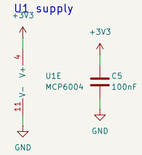
Motor current + sensor Vref (HSTS016L, 0–5 V -> 0–3.21 V, 250 Hz)



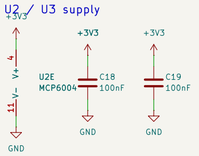
Ground reference to ECU B9 (one point only)



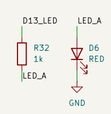
U1 supply



U2 / U3 supply



Status LED (D13)



Power comes from the Feather: 3V3 regulator, USB pin (power-bank 5 V), GND

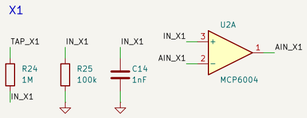


J3: HSTS016L current clamp

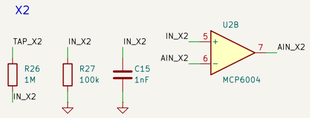


Spare inputs X1–X4 (1 MOhm + 150 kOhm = 0–25 V, MCP6024 buffer -> ADS1115 16-bit, I2C 0x48)

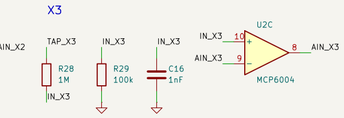
X1



X2

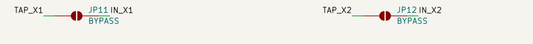


X3

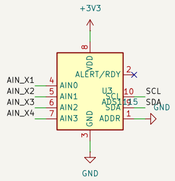


X4





U3: ADS1115



Generated by gen/make_sch.py from gen/design.py
Honda Insight EPS data logger front end

Sheet: /
File: eps_tap_wing.kicad_sch

Title: EPS tap FeatherWing

Size: A2 | Date: 2026-10-26 | **Rev: 1.5**
KiCad E.D.A. 10.0.0 | Id: 1/1